Recommended Assessment
Time of Flight

Exploring Time of Flight Sensors

1. How does the single beam measurements differ between the cardboard and shiny surfaces? When they move slowly? When they oscillate as they get closer?

2. What do you notice about the 3D point cloud? What type of geometric objects can a multizone Time of Flight sensor identify if you're visualizing the 3D point cloud?

3. What do you notice about the depth measurement for the colorized multizone measurement? Is there a limitation to this approach? What happens when the flat surface is placed 1m away from the sensor (and the roll of tape is place 0.5m away)?

Real Life Scenarios

4. You are tasked to select a Time-of-Flight sensor for an aerial system's landing gear. Which type of Time-of-Flight sensor would you pick? a single beam or a multizone sensor?

5. You're working on a manufacturing cell and need to determine if the orientation and distance to incoming parts are correct. How would a multizone Time of Flight sensor be used for this purpose?

Time of Flight Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	